SG Vault Architecture

Strong cryptographic guarantees in a system that fits in your head.

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Mapping the Git Mental Model

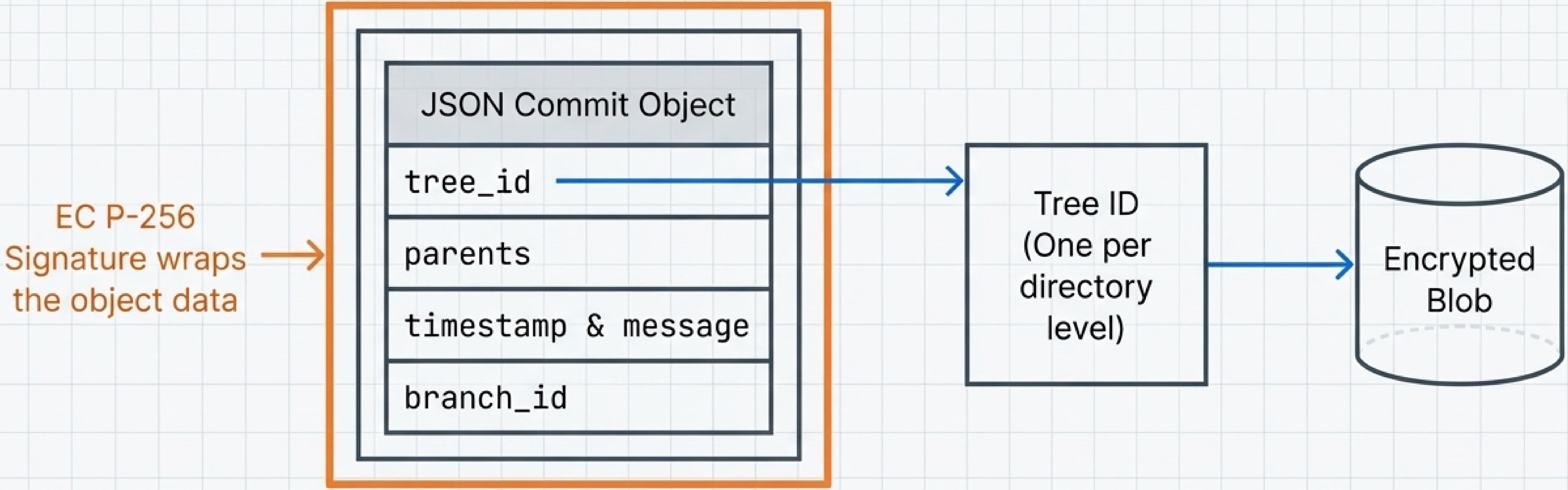
Git Equivalent	SG Vault Primitive	
`git blob`	`blob`	Encrypted file content
`git tree`	`tree`	Encrypted directory listing (one per folder)
`git commit`	`commit`	Signed snapshot with parents, message, timestamp
`git ref`	`ref`	Mutable pointer to a commit (encrypted)
`git branch`	`branch`	Encrypted metadata + key pair identity
[No Git equivalent]	`key`	Public or private key, encrypted in vault

A System Built From Just Six Primitives

blob	Encrypted file content.
tree	Encrypted directory listing (one per folder).
commit	Signed snapshot with parents, message, timestamp.
ref	Mutable pointer to a commit (encrypted).
branch	Encrypted metadata + key pair identity.
key	Public or private key (encrypted in vault).

These live in a flat, folder-grouped storage layout.

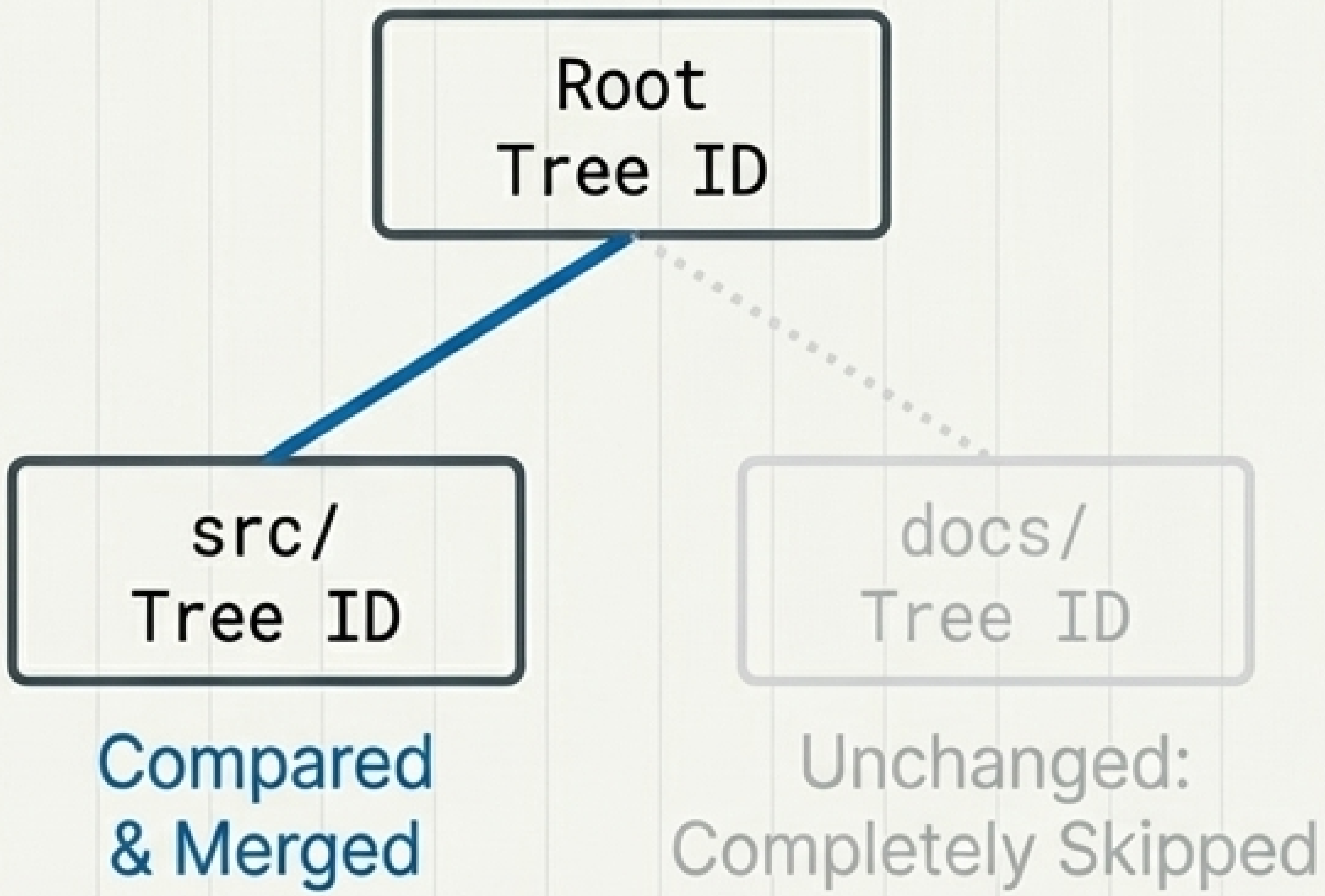
Anatomy of a Signed Snapshot



The object model is pure JSON. You can read a commit object and know what it means without external documentation.

Three-Way Merging Without File Corruption

The Sub-Tree Merge



Compare tree IDs at each directory level.
Efficiency through cryptographic hashing.

Conflict Handling

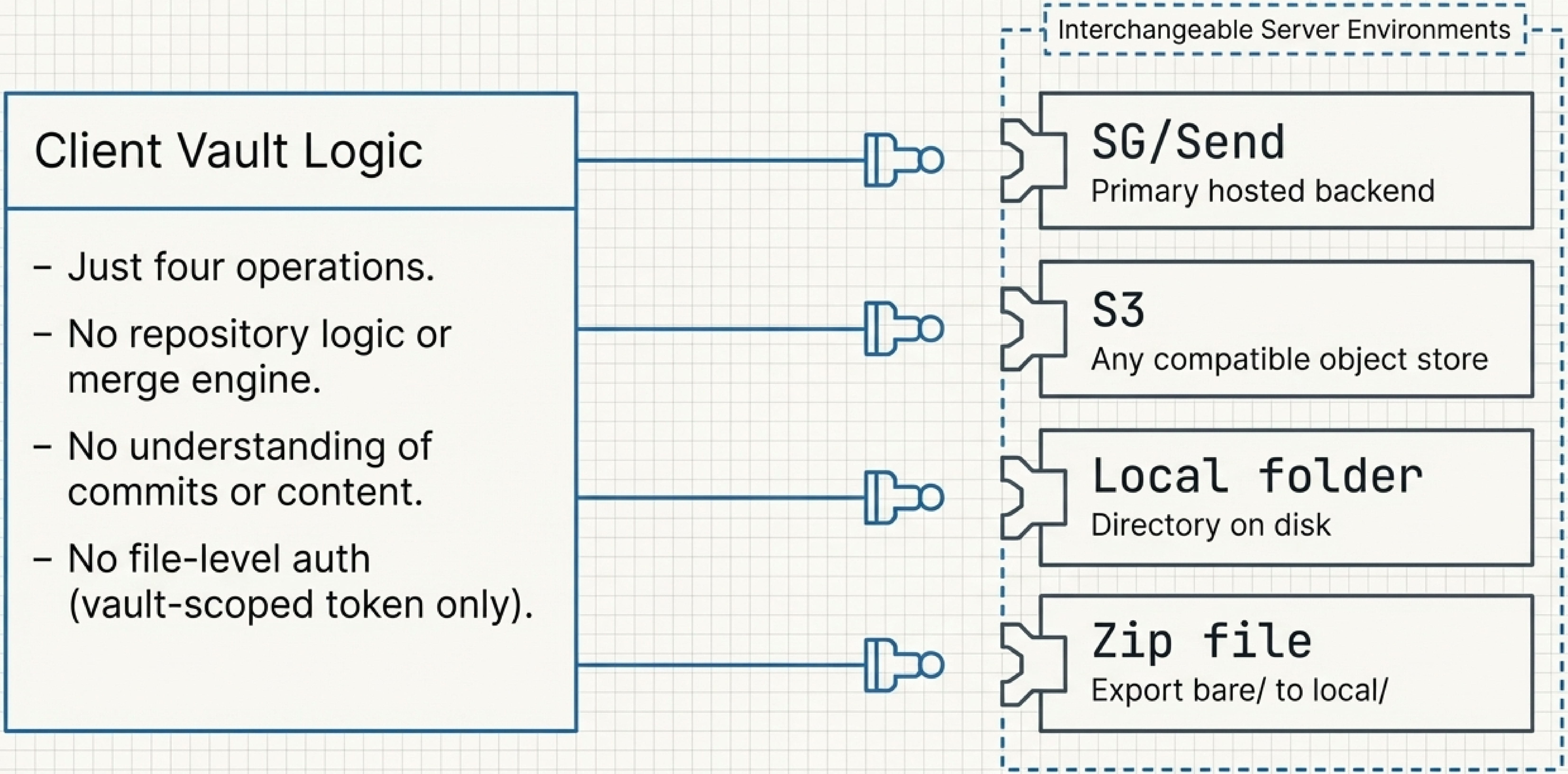
❌ What it isn't

No inline merge markers. No corrupted files injected with conflict strings.

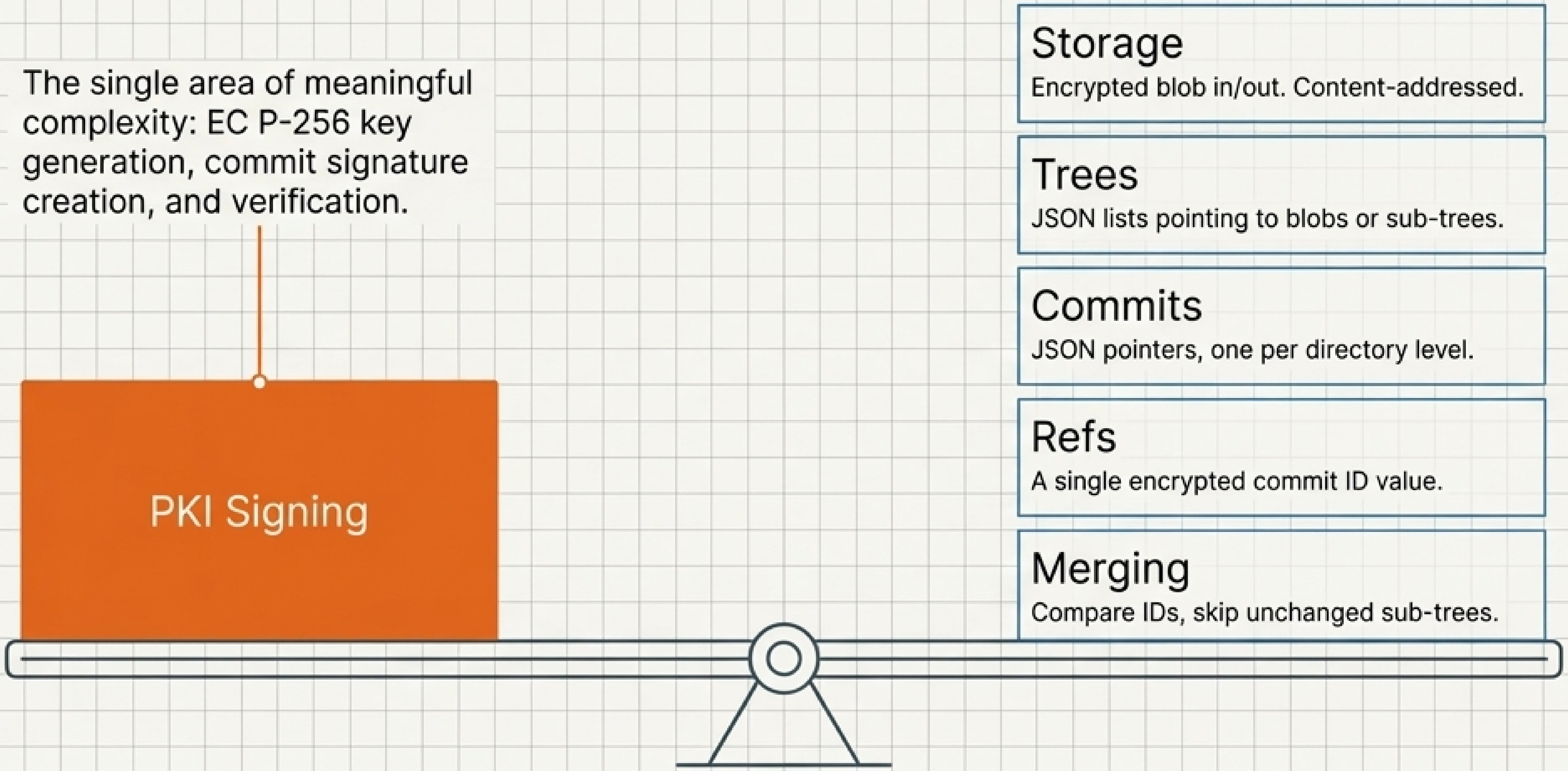
✅ What it is

System safely isolates the collision by generating a discrete `.conflict` file.

The ‘Dumb’ Server Abstraction

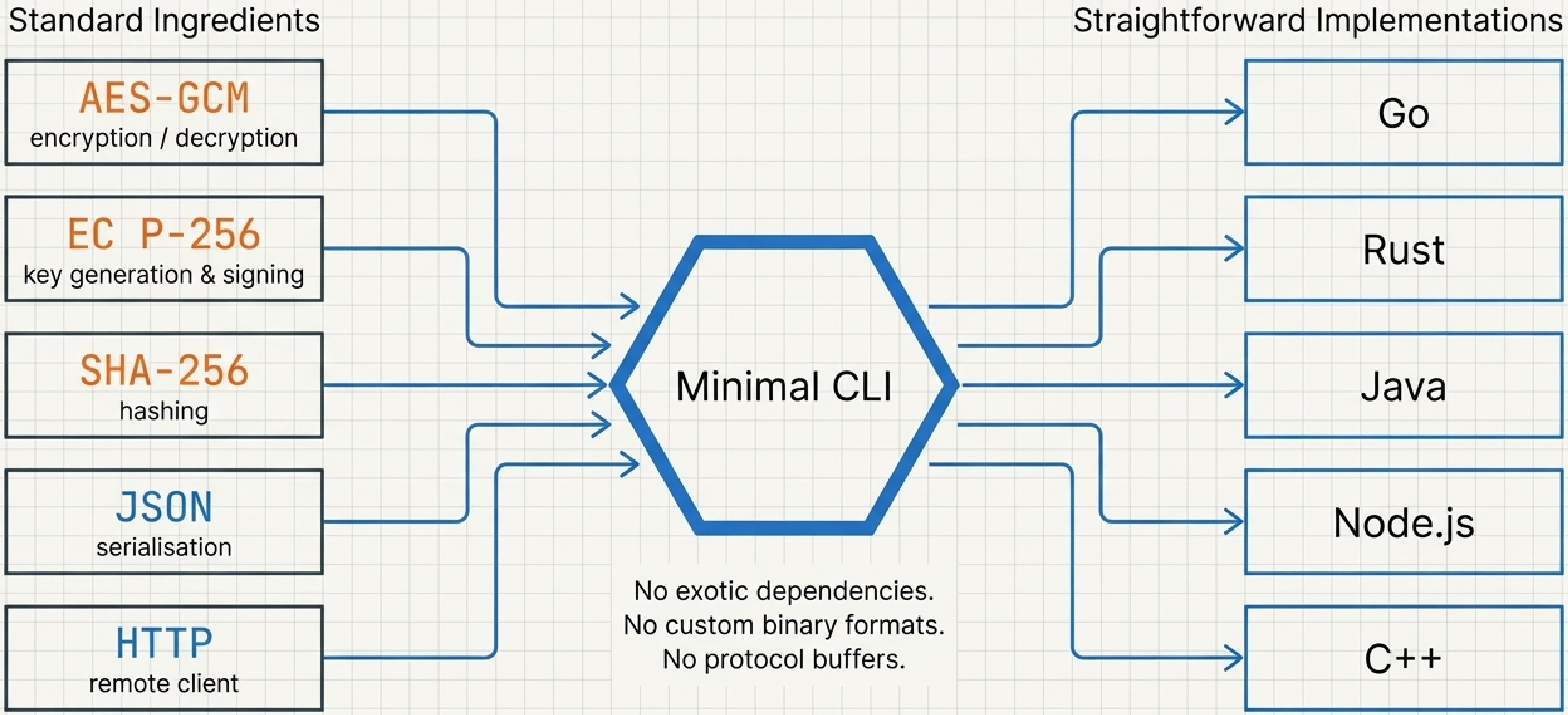


Isolating the Cognitive Load



By isolating cryptographic friction, standard operational patterns remain straightforward.

A Language-Agnostic Recipe



The Equation for Practical Reality

6 Primitives

4 Server Operations

5 Standard Libraries

Auditable
Small primitive count, simple flat-file storage, and JSON objects.

Understandable
Read commit objects instantly without external documentation.

Interoperable
Any client following the schema can read/write the exact same vault.

Easy to Build
Requires only basic cryptography and four storage functions.

The goal: strong cryptographic guarantees with a system that fits in your head.